

# Tarun Mittal

## Senior Software Engineer

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### SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

### SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend and APIs** – FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns, SOA.
- AI and Agents** – LLM agents (OpenAI, Gemini, Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data and Streaming** – Kafka, Flink, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud and Infra** – GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core** – Distributed Systems, Microservices, System Design (HLD/LLD), Multithreading, Concurrency, Caching, Sharding, Mentorship.

### PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

#### Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building AssortSmart behind a **Go (Gin)** write/doing path and planner copilot (**FastAPI, LangGraph, MCP**) so retailers can decide **what to buy, how much, and which stores**, targeting turnaround from days to **under 1 hour** and **≥20 configs** explored per plan.
- Cutting measured clustering-run failures from **8.5% (37/437 on kik)** toward **under 2%** (target) with **14** audited tools that only **read** planning data and **3** human confirm gates before any write-back.
- Building **Hindsight** as the prior-season decision layer with carry-forward and underperformance flags plus Keep/Shop/Drop recommendations, overnight narration **checked against computed metrics**, and tenant metric catalogs that go live **without a code deploy**.
- Keeping plan edits at **~0.4 ms** and month rollups **sub-second** via choice-cluster-week aggregates (**~25M** rows) instead of a flat store-week grid.
- Adopting per-tenant **ClickHouse (63 tables / 8 layers)**, insert-only / partition-swapped, agent read-only) after a row-identical POC cut **250M-row** pivots from **189s to 12.3s** (**~15.5x**, measured).
- Tech** – Go, Gin, Python, FastAPI, LangGraph, MCP, Kafka, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

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UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

#### Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owined design and architecture for Uber's **FRM Risk Scoping** platform (**FastAPI, MySQL, React**) across **8 screens** and **30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation (**~2 weeks to ~3–4 days**).
- Replaced the Google Sheets quarterly close with a **MySQL system of record** that ties **Oracle HFM** extracts to public **10-Q** filings so scoping feeds **PwC** audit work papers with a durable line-level audit trail.
- Automated group and component scoping with residual-risk and EMI workflows that auto-flag material items across **55 financial line items** and **14 entities** against materiality and residual thresholds (**\$340M / \$170M** sample).
- Led 3 engineers** through design reviews, API contracts, and CI quality gates, growing the Bazel pytest suite to **1,100+ tests** for PwC-facing releases.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

#### Menu Ingestion and Automation Platform (Uber Eats)

- Cut menu onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month** by shipping **Selenium** scrapers on GCP with **Kafka** ingest and **Flink** keyed normalize/dedupe and replay.
- Converted unstructured multilingual menus (**PDFs/images**) into Uber Eats schema at **98% fidelity / 100% schema consistency** (offline eval) via a **LangChain RAG** pipeline with **Gemini 2.5 Pro** over a **vector store** of labeled menus, plus supervised fine-tuning.
- Raised successful ingestions to **95%+** by handling anti-bot defenses with IP rotation, dynamic proxy pools, and adaptive backoff.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, vector DB, SFT, GCP, Docker.

#### ANZ Driver Document Compliance (Uber Mobility)

- Cut manual verification by **20 hours/week** by automating **driver and vehicle document compliance** to **99.9%** for **Uber drivers in ANZ** against local authorities.

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MASTERS INDIA, Noida

December 2022 – June 2024

#### GST Compliance and E-Invoicing Platform

- Cut p95 latency from **1.2s to 300ms** for **1,500+ clients** by strangling a PHP (**Laravel**) monolith into **FastAPI microservices** and mentoring **2 engineers**.
- Lifted throughput from **700 to 4,000 requests/min** and processed **1M+ IRP submissions/day (100K+/import)** by building an event-driven path on **Kafka** and **quarter-sharded PostgreSQL**.
- Cut redundant DB reads **30%** and made IRP replay safe by shipping **idempotent sinks (idempotency keys, retries, DLQ)** with Redis cache-aside.
- Cut incident triage **70%** and support tickets **35%** by shipping **ELK / New Relic** on-call alerting plus a client usage dashboard, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

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GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from PHP to **Django (Python)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on MySQL, MongoDB, Redis, and Elasticsearch.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** by orchestrating cron pipelines for video processing, reminders, and recording cleanup.

### EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

### ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications ([HackerRank](#), [LangChain](#)).